

Econometrics for Finance

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September, 2026

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Introduction

Welcome to the course **Econometrics for Finance**.

- My email address: morteza.aghajanzadeh@hhs.se
 - ▶ I will try to answer your questions as soon as possible.
- Three upcoming lectures (depending on class discussion, this may take more or less time)
 - ▶ Covering the basic concepts.
 - ▶ Some parts get very technical, but they are included only for formality.
 - ▶ Keep the main intuition in your mind.
 - ▶ Implement the methods in Python.
- All slides are built on courses I took during my studies.
 - ▶ I have also used slides from previous years.
 - ▶ All errors are mine.
- Office hours for technical questions:
 - ▶ Book a time slot via the [Booking Calendar](#) or [my personal website](#).
 - ▶ Send me your questions by email in advance so I can prepare.
 - ▶ If you cannot find a time slot that suits you, send me an email.

Introduction

In this course, we aim to learn how to use econometrics in finance.

At the end of this course, you should be able to:

- Understand the basic concepts of econometrics.
- Implement the econometric methods.
- Critically evaluate papers.

Important point

I will not grade your thesis, so feel free to ask me anything. Use the office hours to discuss your thesis with me.

Suggested textbooks

- *Introductory Econometrics for Finance* (Brooks (2019))
 - ▶ There is a free [Python Guide](#) for this book (Wichmann and Brooks (2019)).
- *Mostly Harmless Econometrics: An Empiricist's Companion* (Angrist and Pischke (2009))
- You can find more in the uploaded document.

File structure

- Literature
 - ▶ Papers
 - ▶ Reports
 - ▶ ...
- Data
 - ▶ Raw data
 - ▶ Cleaned data
- Code
 - ▶ Split your code into different parts to make it more readable.
 - ▶ Comment your code.
- Meetings
 - ▶ Per meeting summary, notes, and action items.
- Doc
 - ▶ Drafts
 - ▶ Slides
 - ▶ ...
- Output
- README.txt

File structure

Example



code



data



doc



literature



meetings



out



status



README.md

Research question

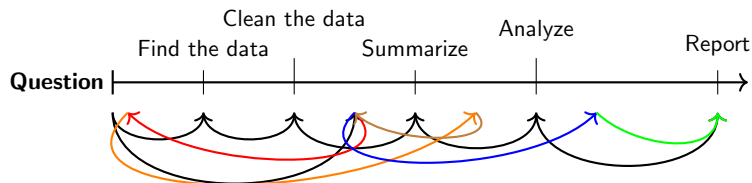
Research pipeline

Frequently asked questions (FAQs):

- ① What is the causal relationship of interest?
- ② What is the ideal experiment that could be used to capture the causal effect of interest?
 - ▶ Beware of FUQs — fundamentally unidentified questions.
- ③ What is your identification strategy?
- ④ What is your mode of statistical inference?

Research process

Research pipeline



How to use AI in your research?

These are only my suggestions, not the official course policy

- You (can) use AI for everything in your research pipeline.
- But **do not** let AI do your research or your **thinking** for you.

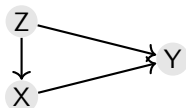
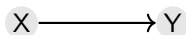
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Causality

- What does causality mean?



- The most important assumption:
 - ▶ a prior ordering of the variables into causes (exogenous variables) and effects (endogenous variables).
- Main problem:
 - ▶ We only have access to observational data.

Some common examples

- Do we really need hospitals?
- Big classes or small classes?
- Years of schooling and wages?
- Unemployment benefits and the unemployment rate?
- ...

The hospitalization question

Group	Sample Size	Mean Health Status	Std. Error
Hospital	7,774	3.21	0.014
No hospital	90,049	3.93	0.003

$$t = \frac{\bar{X}_1 - \bar{X}_0}{\sqrt{SE_1^2 + SE_0^2}} = -50.3$$

Should we conclude that going to hospital makes people less healthy?

- What is the problem?

Potential outcome models

Nowadays, the most common approach to causal inference is the potential outcome model.

- Let's define a binary treatment variable $D_i \in \{0, 1\}$.
- We assume that Y_i is affected by $D_i \Rightarrow Y_i(D_i)$.
- So we have two potential outcomes for each individual:

$$\text{Potential outcome} = \begin{cases} Y_i(1) & \text{if } D_i = 1 \\ Y_i(0) & \text{if } D_i = 0 \end{cases}$$

- So what is the main problem?

Potential outcome models

- We only observe one of the potential outcomes for each individual.

$$\begin{aligned} Y_i &= \begin{cases} Y_i(1) & \text{if } D_i = 1 \\ Y_i(0) & \text{if } D_i = 0 \end{cases} \\ &= Y_i(0) + (Y_i(1) - Y_i(0))D_i \end{aligned}$$

- Stable unit treatment value assumption (SUTVA):
 - ▶ No interference between units.
 - ▶ No hidden versions of treatment.
- We want to know the effect of the treatment on the outcome, defined as:

$$Y_i(1) - Y_i(0)$$

- ▶ And the treatment could have different effects on different individuals.

Average Treatment Effect

Definition

Average Treatment Effect (ATE) is the average of the difference between the potential outcomes for all individuals in the population.

$$ATE = \mathbb{E}[Y_i(1) - Y_i(0)]$$

We still have the same problem:

- We cannot observe $Y_i(1)$ and $Y_i(0)$ for the same individual.

Average Treatment Effect on the Treated

Definition

Average Treatment Effect on the Treated (ATT) is the average of the difference between the potential outcomes for all individuals who receive the treatment.

$$ATT = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1]$$

- We still cannot measure this expectation.
- But now we are close ...
- Let's look at the observed difference in means:
 - ▶ $\mathbb{E}[Y_i \mid D_i = 1]$: average outcome for treated individuals.
 - ▶ $\mathbb{E}[Y_i \mid D_i = 0]$: average outcome for untreated individuals.

Average Treatment Effect on the Treated

$$\underbrace{\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]}_{\text{Observed difference in means}} = \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0]$$
$$= \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 1]$$
$$+ \mathbb{E}[Y_i(0)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0]$$
$$= \text{ATT} + \mathbb{E}[Y_i(0)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0]$$

- Selection bias:

- ▶ The difference in average untreated outcomes $Y_i(0)$ between the treated and the untreated.
- ▶ The apples-and-oranges problem.

- How to eliminate the selection bias?

Assumptions

- Independence (random assignment):

$$\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp D_i \quad \forall i$$

- Conditional independence assumption (CIA):

$$\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp D_i \mid X_i \quad \forall i$$

Key Concepts in Causality

- Endogenous and exogenous variables
- SUTVA
 - ▶ Exclusion restrictions
 - ▶ General/Partial equilibrium effects
- How to eliminate the selection bias?

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Estimation

- We want to estimate the effect of a variable x on a variable y .

$$y_i = g(X_i) + \varepsilon_i$$

- What does it mean?
 - ▶ We want to estimate the function $g(\cdot)$.
 - ▶ Each estimation method has its own assumptions to estimate the function $g(\cdot)$.
 - ▶ Also, we need to make an assumption about the distribution of ε_i .

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Ordinary Least Squares (OLS)

The most common method to estimate the function $g(\cdot)$ is OLS.

Assumptions:

- $g(\cdot)$ is linear in parameters.
- $\mathbb{E}[\varepsilon_i | X_i] = 0$.
- Estimation: $(\beta_{k \times 1}, X'_{i,1 \times k})$

$$Y_i = X_i' \beta + \varepsilon_i \Rightarrow \beta = \arg \min_{\beta} \mathbb{E}[(Y_i - X_i' \beta)^2]$$

$$\text{FOC: } \mathbb{E}[X_i(Y_i - X_i' \beta)] = 0 \Rightarrow \beta = \mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i Y_i]$$

Ordinary Least Squares (OLS)

- Other formulas for β exist, but they are all equivalent (e.g., with a single regressor, $\beta = \frac{\text{Cov}(X_i, Y_i)}{\text{Var}(X_i)}$).
- We can extend it to the case of multiple variables. Stacking all n observations, the sample estimator is

$$Y = X\beta + \varepsilon \Rightarrow \hat{\beta} = (X'X)^{-1}X'Y$$

$$\text{where } \beta_{k \times 1}, \quad X_{n \times k} = \begin{bmatrix} X'_1 \\ \vdots \\ X'_n \end{bmatrix}_{n \times k}, \quad \text{and } Y_{n \times 1}.$$

Let's skip the technical details and focus on the intuition.

Conditional Expectation Function (CEF)

- What is the expected value of Y given X ?

$$\mathbb{E}[Y|X] = \mathbb{E}[Y_i|X_i] = m(X_i)$$

- Linear CEF: $m(X_i) = X_i'\beta$.
- Let's define the **best linear predictor** (the linear projection of Y_i on X_i):

$$\mathbb{E}^*[Y_i|X_i] = X_i'\beta_{OLS}$$

Theorem

Suppose the CEF is linear. Then

$$\mathbb{E}[Y_i|X_i] = \mathbb{E}^*[Y_i|X_i]$$

OLS vs. CEF

- OLS works well when the CEF is linear: the two coincide.
- And when the CEF is *not* linear, OLS still gives the best linear approximation to it.

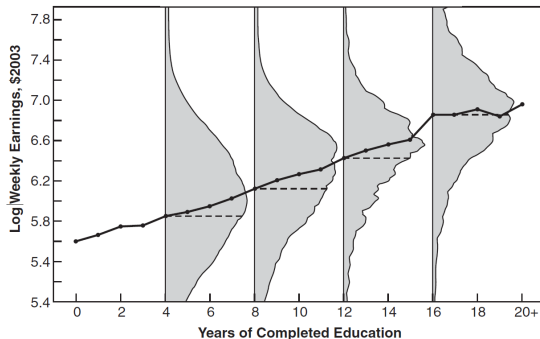


Figure 1: The OLS line and the CEF.

Interpretation of regressions

- Empirical papers until roughly the 1990s (probabilistic causality):
 - ▶ The regression was motivated by a complex model, and the equation was meant to describe an economic process.
 - ▶ All regressors on the right-hand side were more or less equally important.
- Empirical papers nowadays (counterfactual causality):
 - ▶ Focusing on one causal relationship.
 - ▶ Identification strategy meant to eliminate selection bias.
 - ▶ Regressors are not equally important, and only one variable is seen as having causal effects.
 - ▶ All others are controls included in service of the focused causal agenda.

Reading the regression results

- We are interested in the relationship between wages, gender, and race.
- We have the following variables (a binary coding is a strong simplification of the actual data):

$$\text{Gender}_i = \begin{cases} 1 & \text{if Female} \\ 0 & \text{if Male} \end{cases} \quad \text{Race}_i = \begin{cases} 1 & \text{if White} \\ 0 & \text{if Black} \end{cases}$$

- Let's estimate:

$$\widehat{\text{Wage}}_i = \alpha + \beta_1 \times \text{Gender}_i + \beta_2 \times \text{Race}_i + \beta_3 \times \text{Gender}_i \times \text{Race}_i + \varepsilon_i$$

- We have the results:

$$\widehat{\text{Wage}}_i = 10 - 5 \times \text{Gender}_i + 3 \times \text{Race}_i - 2 \times \text{Gender}_i \times \text{Race}_i$$

- What is the interpretation?

Main effect and interaction term

Let's assume that we have the following regression:

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} x_{2i} + \varepsilon_i$$

- $\mathbb{E}[y_i | x_{1i} = 0, x_{2i} = 0] = \alpha$
- $\mathbb{E}[y_i | x_{1i} = 1, x_{2i} = 0] = \alpha + \beta_1$
- $\mathbb{E}[y_i | x_{1i} = 0, x_{2i} = 1] = \alpha + \beta_2$
- $\mathbb{E}[y_i | x_{1i} = 1, x_{2i} = 1] = \alpha + \beta_1 + \beta_2 + \beta_3$

Note: If we include an interaction term, we should also include both main effects.

The saturated model

If x_1 is categorical rather than binary, we can saturate the model in x_1 :

$$y_i = \alpha + \sum_{j=1}^k \beta_j d_{ji} + \delta x_{2i} + \sum_{j=1}^k \gamma_j d_{ji} \times x_{2i} + \varepsilon_i$$

where x_1 takes $k + 1$ values and $d_{1i}, \dots, d_{ki}$ are dummies for those values, with one category omitted as the reference.

- The saturated model imposes no functional form on x_1 : it fits the conditional mean in every cell exactly.

Topics we will not cover

But you should know that they exist

- Frisch-Waugh-Lovell theorem
- R^2
- Weighting regressions
- Weighted least squares (WLS)
- Constraining coefficients (CLS)
- Maximum likelihood estimation (MLE)
- Generalized method of moments (GMM)
- Logit and probit models
- Normalization of variables
- ...

Questions?

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Asymptotic Properties of OLS

- We have estimated the model, which means that we have the β .
- How can we know that the β is the correct answer?
- What do we mean by the correct answer?
- What is the difference between the true β and the estimated β ?
- What is the distribution of the estimated β ?

Population parameter vs. Sample estimator

What is the difference between :

- Population parameter: β
- Sample estimator: $\hat{\beta}$

Let's estimate the average height of the students in SSE.

- Population parameter: The average height of all students in SSE.
- Sample estimator: The average height of the students in each class.
 - ▶ Other sample estimators:
 - ★ The average height of the students in The Atrium.
 - ★ The height of the first student who enters the class.
 - ★ The maximum height of the students who pass the main entrance.
 - ★ ...

We have one true β (The average height of all students in SSE) and many sample estimators (The average height of the students in different places).

Unbiased Estimators

Definition (Unbiased Estimator)

An estimator $\hat{\theta}$ is *unbiased* for a parameter θ if $E(\hat{\theta}) = \theta$ for all θ in the parameter space.

- An unbiased estimator gives us the correct answer on average over many experiments.
- In our example, it means that if we repeat the experiment many times with other classes, the average of the estimates will equal the average height of all students in SSE.

Consistent Estimators

Definition (Consistent Estimator)

An estimator $\hat{\theta}_n$ is *consistent* for a parameter θ if $\hat{\theta}_n$ converges in probability to θ as the sample size n grows without bound.

- A consistent estimator gets closer and closer to the true parameter as the sample size increases.
- In our example, it means that as we enlarge the sample – from the students in this class, to all classes on this floor, to the whole school – our estimate gets arbitrarily close to the average height of all students in SSE.

Data Generating Process

Definition (Data Generating Process)

The Data Generating Process (DGP) refers to the underlying, often unknown, mechanism that generates the observed data. // The DGP embodies the true relationship among variables in a dataset.

- The DGP is the “ground truth”, which we often aim to discover or approximate in statistical modeling and analysis.
- While we typically cannot know the exact DGP, we can use statistical models to make educated guesses or approximations about it.
- Understanding the DGP is crucial for model specification, hypothesis testing, estimation, and prediction.

OLS estimator

$\hat{\beta}$

- 1 **Linear in parameters:** $y_i = \alpha + \sum_{k=1}^K \beta_k x_{ki} + \varepsilon_i$
- 2 **Random Sampling:** (y_i, x_i) are i.i.d. draws from a joint distribution F .
- 3 **No perfect collinearity:** $\text{rank}(\mathbb{E}[X_i X_i']) = K$.
- 4 **Zero conditional mean:** $\mathbb{E}[\varepsilon_i | X_i] = 0 \Rightarrow \text{cov}(\varepsilon_i, x_i) = 0$.

Theorem

If the above assumptions hold, then the OLS estimator is an unbiased estimator of β .

Proof.

$$\mathbb{E}[\hat{\beta}] = \mathbb{E}[(X'X)^{-1}X'Y] = \beta + \mathbb{E}[(X'X)^{-1}X'\varepsilon] = \beta$$



Biased estimator

- What would happen if we include irrelevant variables?
- What would happen if we exclude relevant variables?
- Endogeneity.

OLS estimator

$$\text{Var}(\hat{\beta})$$

Recall that:

$$\hat{\beta} = (X'X)^{-1}X'Y = \beta + (X'X)^{-1}X'\varepsilon$$

$$\text{Var}(\hat{\beta}) = \text{Var}((X'X)^{-1}X'\varepsilon) = (X'X)^{-1}X'\text{Var}(\varepsilon)X(X'X)^{-1}$$

⑤ **Homoskedasticity:** $\text{Var}(\varepsilon|X) = \sigma^2 I_n$.

Theorem

If the above assumptions hold, then the variance of OLS estimator is:

$$\text{Var}(\hat{\beta}) = \sigma^2(X'X)^{-1}$$

OLS estimator

$$\text{Var}(\hat{\beta})$$

- How can we estimate σ^2 ?
- Heteroskedasticity
 - ▶ White standard errors
 - ▶ Clustered standard errors
 - ▶ Bootstrap standard errors
 - ▶ ...

Next step

- Now we have estimated the model, which means that we know $\hat{\beta}$.
- Each coefficient has its own standard error.
- How can we test the significance of each coefficient?
- What do we mean by the significance of each coefficient?

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Hypothesis Testing

Review from Statistics

- **Null Hypothesis (H_0):** A statement about the value of a population parameter. This is the hypothesis we are testing. Usually, it involves a statement of “no effect” or “no difference”.
 - ▶ Example: “Average height of students in SSE is 170 cm.”
 - ▶ Formal Example: $\mu = \mu_0, \sigma = \sigma_0$
- **Alternative Hypothesis (H_1 or H_a):** The opposite of the null hypothesis. It is what we believe to be true if we reject the null hypothesis.
 - ▶ Example: “Average height of students in SSE is not 170 cm.”
 - ▶ Formal Example: $\mu \neq \mu_0, \sigma > \sigma_0$

Hypothesis Testing

Review from Statistics

- **Test Statistic:** A numerical summary of the data that is used in making the decision about the rejection of the null hypothesis.
- **Rejection Region or Critical Region:** The values of the test statistic for which we reject the null hypothesis.
- **P-value:** The probability, under the null hypothesis, of obtaining a value of the test statistic as extreme or more extreme than the one computed from the sample data.

Understanding the P-value

- **P-value:** The p-value measures how compatible the observed data are with the null hypothesis. It does *not* tell us whether a hypothesis is true.
- It is a number between 0 and 1 and is interpreted in the following way:
 - ▶ A small p-value (typically ≤ 0.05) indicates strong evidence against the null hypothesis, so you **reject the null hypothesis**.
 - ▶ A large p-value (> 0.05) indicates weak evidence against the null hypothesis, so you **fail to reject the null hypothesis**.
 - ▶ P-values very close to the cutoff (0.05) are considered to be marginal (could go either way).
- Always report the p-value so your readers can draw their own conclusions.

Hypothesis Testing in Regression

- Recall that:

- ▶ $\hat{\beta} = \beta + (X'X)^{-1}X'\varepsilon$
- ▶ $\text{Var}(\hat{\beta}) = \sigma^2(X'X)^{-1}$
- ▶ $\mathbb{E}[\hat{\beta}] = \beta$

- What do we need to know for testing the hypothesis?

- 6 **Normality:** $\varepsilon \sim N(0, \sigma^2)$.

Theorem

Under the six classical linear model (CLM) assumptions, we have:

$$\hat{\beta} \sim N(\beta, \text{Var}(\hat{\beta}))$$

Hypothesis Testing in Regression

t-test

- **Hypothesis:**

$$\begin{cases} H_0 : \beta_k = 0 \\ H_1 : \beta_k \neq 0 \end{cases}$$

Theorem

Under the CLM assumptions, we have:

$$t = \frac{\hat{\beta}_k - 0}{SE(\hat{\beta}_k)} \sim t_{n-K-1}$$

- $SE(\hat{\beta}_k)$ is the standard error of $\hat{\beta}_k$.
- t_{n-K-1} is the t-distribution with $n - K - 1$ degrees of freedom.
- We reject the null hypothesis if $|t| > t_{n-K-1}(\frac{\alpha}{2})$.
- **We want to reject the null hypothesis!**

Test interpretation

Be careful about your interpretation after the test!

It is true that we reject the null hypothesis, but we cannot say that the null hypothesis is false.

What can we say?

We only can say that we have (or do not have) enough evidence to reject the null hypothesis.

Other concepts

- F-test
- Why do we care about the standard error of $\hat{\beta}$?
- Confidence interval

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Identification Arsenal

Not estimation anymore

- The estimation of the model is not a problem anymore.
- **Identification:** Are we really estimating the model that we want to estimate? Or are we estimating something else?



Identification Arsenal

Recap

Our arsenal

- Use **Matching** to create balance on observables between the treatment and control groups.
- Use **Instrumental Variable (IV)** to isolate exogenous variation in the treatment.
- Use **Panel Data** or **Repeated Cross-section** to control for unobserved heterogeneity:
 - ▶ **Difference-in-Differences (DiD)**
 - ▶ **Synthetic Controls (SC)**
 - ▶ **Event study**
- Use **Regression Discontinuity Design (RDD)** to exploit exogenous variation in the treatment around a cutoff.

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Panel Data

- **Panel data** is a combination of cross-sectional and time-series data.
- We have at least two dimensions:
 - 1 i : N individuals
 - 2 t : T periods
- What is the difference between panel data and time-series data, or repeated cross-section data?
- Panel data can be **balanced** or **unbalanced**. We have to be careful about this.
 - ▶ If the panel is unbalanced, missing observations should be random, i.e., unrelated to the error term μ_{it} .

Advantages of Panel Data

- **Controlling for Individual Heterogeneity:** Panel data allow us to control for time-invariant variables we cannot observe or measure.
- **Temporal Dynamics:** Panel data can help us study the impact of changes within individuals over time.
- **More Data Points:** With both cross-sectional and time-series dimensions, panel data provide more data points, increasing degrees of freedom and reducing collinearity among explanatory variables.
- **Identifying Effects:** Panel data enable the identification and measurement of effects that are simply not detectable in pure cross-section or time-series data.

Example of Panel Data

Wide Format

Team	Points	Assists	Rebounds
A	88	12	22
B	91	17	28
C	99	24	30
D	94	28	31

Long Format

Team	Variable	Value
A	Points	88
A	Assists	12
A	Rebounds	22
B	Points	91
B	Assists	17
B	Rebounds	28
C	Points	99
C	Assists	24
C	Rebounds	30
D	Points	94
D	Assists	28
D	Rebounds	31

Panel Data Intuition

Suppose that the DGP is:

$$Y_{it} = \alpha + \beta x_{it} + \varepsilon_{it}$$

What do you know about ε_{it} ?

- Possible cases:
 - ▶ $\varepsilon_{it} = a_i + \mu_{it}$
 - ▶ $\varepsilon_{it} = a_i + b_t + \mu_{it}$
- What do you think about a_i ?
- Let's keep it simple and assume that the first case is true.

Panel Data Intuition

a_i is the **individual effect** or **individual heterogeneity**. (Examples: **ability, talent, intelligence, etc.** if we are studying the effect of mandatory training for unemployed individuals on their future wages)

- ① a_i is random with variance σ_a^2 and independent of x_{it} . → **No OVB**
(Example: the number of individual training hours is not correlated with the ability of the individual)
- ② a_i is correlated with x_{it} . → **OVB**
(Example: education level is correlated with the ability of the individual)

This is the first assumption we need to make with panel data, but we still have to be careful.

Independent individual effect

$$\text{Cov}(a_i, x_{it}) = 0$$

If a_i is random and independent of x_{it} , then we do not have the OVB problem.

Is there any problem?!

We have a serial correlation problem: for the same individual in two different periods $t \neq s$,

$$\mathbb{E}[\varepsilon_{it}\varepsilon_{is}] = \mathbb{E}[(a_i + \mu_{it})(a_i + \mu_{is})] = \mathbb{E}[a_i^2] = \sigma_a^2 \neq 0, \quad \text{Var}(\varepsilon_{it}) = \sigma_a^2 + \sigma_\mu^2$$

$$\mathbb{E}[\varepsilon\varepsilon'] = \Omega = \begin{bmatrix} \sigma_a^2 + \sigma_\mu^2 & \cdots & \sigma_a^2 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \sigma_a^2 & \cdots & \sigma_a^2 + \sigma_\mu^2 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & \sigma_a^2 + \sigma_\mu^2 & \cdots & \sigma_a^2 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \sigma_a^2 & \cdots & \sigma_a^2 + \sigma_\mu^2 \end{bmatrix}$$

Independent individual effect

$$\text{Cov}(a_i, x_{it}) = 0$$

1 Pooled OLS (POLS):

- ▶ Simply ignore the panel structure and run OLS on the pooled data.
- ▶ We have to be careful about the standard errors.
- ▶ We need to cluster the standard errors at the individual level.

2 Random Effects (RE):

- ▶ Uses (feasible) GLS, exploiting the structure of the error covariance matrix Ω .
- ▶ More efficient than POLS.

Note: Recall the main assumption

- a_i is random and independent of x_{it} .

Dependent individual effect

$$\text{Cov}(a_i, x_{it}) \neq 0$$

So, we have the omitted variable bias problem.

What is the solution?

- Find a proxy for a_i .
- Find an instrument for x_{it} that is uncorrelated with a_i .
- Randomize the treatment.

Is there any other solution? **Hint:** What additional features does panel data provide that are not available in cross-sectional data?

- We can control for a_i . This is the **Fixed Effects (FE) model**.

Fixed Effects model

Assumption

$$y_{it} = \alpha + \beta x_{it} + a_i + \mu_{it}$$

- We need to assume that the regressors in every period are uncorrelated with the error term in every period (**strict exogeneity**)¹:

$$\mathbb{E}(x_{is}\mu_{it}) = 0, \quad \forall s, t \in \{1, \dots, T\}$$

¹If it does not hold (e.g., with a lagged dependent variable), you can use the **Arellano–Bond** estimator.

Fixed Effects model

Estimation

There are three ways to estimate the FE model:

- ① **Direct estimator (LSDV)**: Add a dummy variable for each individual and run OLS on this augmented regression.
- ② **Within estimator**: Subtract each individual's time average from each variable and then run OLS on the transformed data.
- ③ **First-difference estimator**: Take the first difference of each variable and then run OLS on the transformed data. (I do not cover this one.)

Direct estimator

We add a dummy variable for each individual and run OLS on this augmented regression.

This helps to understand the intuition behind the FE model.

Write the model as:

$$y_{it} = \alpha + \beta x_{it} + \sum_{j=2}^N a_j d_{j,i} + \mu_{it}$$

- $d_{j,i}$ is a dummy variable for individual j .
- $d_{j,i} = 1$ if observation i belongs to individual j and $d_{j,i} = 0$ otherwise.
- One dummy is dropped because we keep the constant α (dummy variable trap).

With a large number of individuals, we have many dummy variables, which makes estimation costly.

Within estimator

Recall that our data has two dimensions: i and t .

$$y_{it} = \alpha + \beta x_{it} + a_i + \mu_{it}$$

- ① Average over time $t = 1, \dots, T$ ($\bar{x}_i = \frac{1}{T} \sum_t x_{it}$)

$$\bar{y}_i = \alpha + \beta \bar{x}_i + a_i + \bar{\mu}_i$$

- ② Subtract the time average from the original data

$$(y_{it} - \bar{y}_i) = \beta(x_{it} - \bar{x}_i) + (\mu_{it} - \bar{\mu}_i)$$

- ③ Define $\tilde{y}_{it} = y_{it} - \bar{y}_i$ and $\tilde{x}_{it} = x_{it} - \bar{x}_i$ and then run OLS on the transformed data.

The within estimator gives numerically the same $\hat{\beta}$ as the direct estimator (FWL theorem), but it remains feasible with a large number of individuals. (If you demean manually, the standard errors need a degrees-of-freedom correction.)

Choosing between Fixed Effects (FE) and Random Effects (RE)

- **Fixed Effects (FE):**

- ▶ Uses only within-unit variation over time.
- ▶ Removes all time-invariant unobserved heterogeneity (but not time-varying confounders).
- ▶ Good when you believe differences across units (e.g., individuals, firms) might be correlated with the independent variables.

- **Random Effects (RE):**

- ▶ Treats the individual effect a_i as random.
- ▶ Suitable when you believe the variability across units is uncorrelated with the independent variables.
- ▶ More efficient than FE, but inconsistent if this assumption fails.

- **Hausman Test:**

- ▶ A statistical test to decide between FE and RE.
- ▶ Null hypothesis: The difference in coefficients is not systematic.
- ▶ If significant, prefer FE over RE.

Practical issue

In addition to individual fixed effects, we often need to control for common shocks over time:

$$y_{it} = \alpha + a_i + \beta x_{it} + \mu_{it}$$

$$y_{it} = \alpha + a_i + b_t + \beta x_{it} + \mu_{it}$$

Definition: b_t is the **time effect** or **time heterogeneity**.

- $a_i = a_1 D_{i=1} + \cdots + a_{N-1} D_{i=N-1}$
- $b_t = b_1 D_{t=1} + \cdots + b_{T-1} D_{t=T-1}$

Coefficients intuition

Be careful about the interpretation of the coefficients.

- Think of running a separate regression for each individual and then taking a weighted average of the slopes, with weights proportional to the within-individual variance of x_{it} . (This is just an intuition, not exactly what is going on.)
- We only get the average effect of x_{it} on y_{it} .

Understanding the Heterogeneity

A puzzle

Example: Income and Access to Clean Water

- Panel of households i over years t .
- Question: what is the effect of income (x_{it}) on access to clean water (y_{it})?

We run two regressions:

- 1 **Pooled OLS:** $y_{it} = \alpha + \beta x_{it} + \varepsilon_{it} \rightarrow$ large, significant $\hat{\beta}$
- 2 **Household FE:** $y_{it} = a_i + \beta x_{it} + \mu_{it} \rightarrow \hat{\beta} \approx 0$, not significant

What do you expect to see in the FE model?

FE absorbs all the differences between richer and poorer households, so it may find no effect.

Understanding the Heterogeneity

What happened?

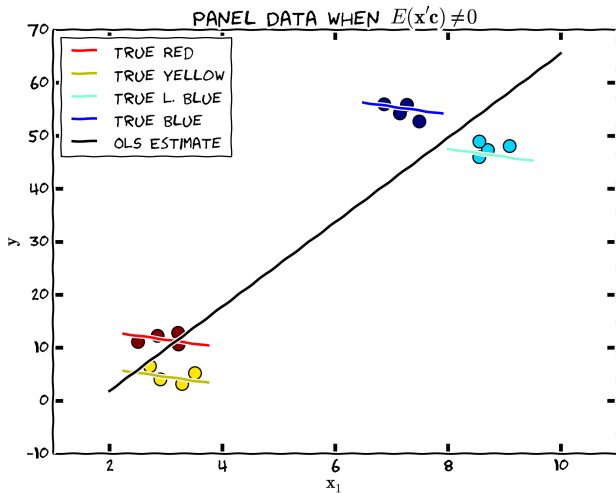
Ask: **which variation** in income does each estimator use?

- **Pooled OLS** compares *different* households: richer vs. poorer.
 - ▶ Higher-income households generally have better access (cross-sectional relationship).
- **FE** compares the *same* household over time: it removes all differences between richer and poorer households.
 - ▶ Within-household changes in income are small and do not significantly alter access.
- So FE may find no effect because:
 - ▶ within-household variation is too small to detect it (low power), or
 - ▶ FE identifies a different, short-run effect.

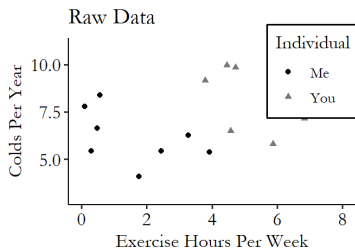
Warning

Sometimes the effect you care about lives in the cross-sectional dimension, and by using the FE model, you may lose it. Be careful about the underlying mechanisms (and remember that cross-sectional correlations can also be confounded).

Intuition



Intuition



Assignment I

- 1 The dataset “local_returns.dta” contains panel information on the monthly stock returns of firms (variable: `ret`), returns of firms in the same area (variable: `city_returns`), returns of firms in the same industry (variable: `indret`) and a few other risk factors (MRP, HML, MOM) from 2000 to 2010. The variable `permno` is the firm identifier, while `date` is the time variable. Individual firms (i.e., `permnos`) are allocated to 20 cities and 12 industries.
 - a Is it a balanced or unbalanced panel? Provide aggregate summary statistics.
 - b Report the average return and volatility of firms in the 20 cities (i.e., summary statistics by city). Do you see any differential returns among cities?
 - c We are interested in the effect of local returns on firm returns. Our model is:

$$\text{ret}_{it} = \alpha + \beta_1 \text{city_returns}_{it} + \beta_2 \text{indret}_{it} + \varepsilon_{it}$$

Estimate the model using pooled OLS and interpret the results. Test for heteroskedasticity in the residuals.

- d Which dimension should be used as a fixed effect? Estimate the model using your proposed fixed effects and interpret the results.
- e Let's try two-way fixed effects. Estimate the model using two-way (firm and time) fixed effects and interpret the results.

Assignment II

- Re-estimate the two-way fixed effects model with standard errors double-clustered by firm and time. Compare your results with the other estimations.

Extra Assignment

A Passage to America: University Funding and International Students

- ① What is the research question?
- ② What is the identification strategy?
- ③ What is the main result?
- ④ Try to replicate Table 3 (OLS estimates only; the rest of the table is optional).

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Motivation

- What is panel data?
- Is it easy to find panel data?
- What if there is no panel data, or treatment is group-level?
 - ▶ Effect of a policy in a specific location
 - ▶ Effect of the minimum wage on employment in different industries
 - ▶ Effect of contaminated drinking water on cholera epidemics
- **Treatment** is assigned at the group level (region, industry).
- We may not observe the same individuals over time.
- We need a **control group** for comparison. How?!
- Build a group-by-period panel from repeated cross-sections.
 - ▶ This controls for time-invariant group-level heterogeneity.

DiD vs. FE

$$y_{it} = a_i + b_t + \beta D_{it} + \varepsilon_{it}$$

Assume that we have a binary treatment variable D_{it} , which is equal to 1 if individual i is treated at time t . Potential outcomes from the FE point of view:

$$\mathbb{E}[y_{it}^0|i, t] = a_i + b_t, \quad \mathbb{E}[y_{it}^1|i, t] = a_i + b_t + \beta$$

From the DiD point of view, we divide units into groups s , and the treatment is assigned at the group level:

$$\mathbb{E}[y_{ist}^0|s, t] = c_s + b_t, \quad \mathbb{E}[y_{ist}^1|s, t] = c_s + b_t + \beta$$

The basic strategy is the same as FE, but we don't need repeated observations of each individual. We can use repeated cross-section data to build a group-by-period panel.

Difference-in-Differences

Simple intuition

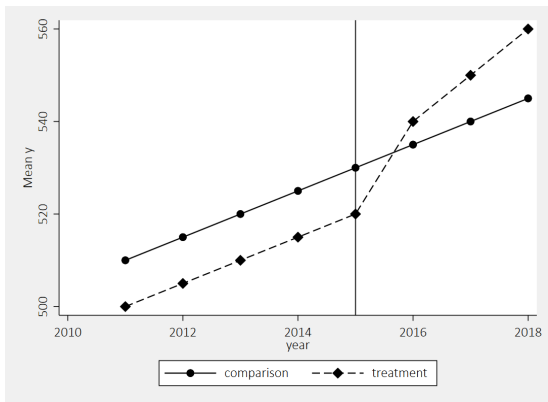
Compare the change in average outcomes of treated vs. control groups:

- **Trend**: broad economic changes (common to both groups)
- **Effect**: treatment effect

	Before	After	After – Before
Control (C)	C_1	$C_1 + \text{Trend}$	Trend
Treated (T)	T_1	$T_1 + \text{Trend} + \text{Effect}$	Trend + Effect
Treated – Control:			Effect

Key assumption (parallel trends): without treatment, T changes like C .

Example



Identification Arsenal

Recap

Our arsenal

- Use **Matching** to create balance between treatment and control group.
- Use **Instrumental Variable (IV)** to create exogenous variation in the treatment.
- Use **Panel Data** or **Repeated Cross-section** to control for unobserved heterogeneity:
 - ▶ **Difference-in-Differences (DD)**
 - ▶ **Synthetic Controls (SC)**
 - ▶ **Event study**
- Use **Regression Discontinuity Design (RDD)** to create exogenous variation in the treatment.

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Difference-in-Difference

Estimation

Estimation:

$$y_{ist} = \alpha + \gamma_s + \lambda_t + \beta D_{ist} + \varepsilon_{ist} \quad \rightarrow \beta = \text{treatment effect}$$

where

- y_{ist} is the outcome of interest for individual i in group s at time t
- γ_s is the group fixed effect
- λ_t is the time fixed effect
- D_{ist} is the treatment dummy which is 1 for treated group after the treatment

$$\begin{aligned} C_1 &= \alpha, & C_2 &= \alpha + \lambda, & T_1 &= \alpha + \gamma_s, & T_2 &= \alpha + \gamma_s + \lambda + \beta \\ \Rightarrow (T_2 - T_1) - (C_2 - C_1) &= \beta \end{aligned}$$

Difference-in-Difference

Another point of view (Additional concepts)

- Consider the following structure (block structure):

$$\mathbf{D} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

- Define the CEF matrix:

$$\begin{bmatrix} \mathbb{E}[Y_{i0}|D_i = 0] & \mathbb{E}[Y_{i1}|D_i = 0] \\ \mathbb{E}[Y_{i0}|D_i = 1] & \mathbb{E}[Y_{i1}|D_i = 1] \end{bmatrix}$$

- The Potential Outcome Matrix (POM) for $d \in \{0, 1\}$:

$$\begin{bmatrix} \mathbb{E}[Y_{i0}(d)|D_i = 0] & \mathbb{E}[Y_{i1}(d)|D_i = 0] \\ \mathbb{E}[Y_{i0}(d)|D_i = 1] & \mathbb{E}[Y_{i1}(d)|D_i = 1] \end{bmatrix}$$

Potential Outcome Matrix

Additional concepts

- Elements of POM could be **counterfactual** or **factual**
- We have eight Potential outcomes in the 2×2 block structure:

$$Y_{it}(0) : \begin{bmatrix} \mathbb{E}[Y_{i0}(0)|D_i = 0] & \mathbb{E}[Y_{i1}(0)|D_i = 0] \\ \mathbb{E}[Y_{i0}(0)|D_i = 1] & \mathbb{E}[Y_{i1}(0)|D_i = 1] \end{bmatrix}$$

$$Y_{it}(1) : \begin{bmatrix} \mathbb{E}[Y_{i0}(1)|D_i = 0] & \mathbb{E}[Y_{i1}(1)|D_i = 0] \\ \mathbb{E}[Y_{i0}(1)|D_i = 1] & \mathbb{E}[Y_{i1}(1)|D_i = 1] \end{bmatrix}$$

- So the CEF matrix corresponds to the factual potential outcomes:

$$Y_{it} : \begin{bmatrix} \mathbb{E}[Y_{i0}(0)|D_i = 0] & \mathbb{E}[Y_{i1}(0)|D_i = 0] \\ \mathbb{E}[Y_{i0}(0)|D_i = 1] & \mathbb{E}[Y_{i1}(1)|D_i = 1] \end{bmatrix}$$

Identification

Additional concepts

- We are interested in the treatment effect:

$$\begin{aligned}\tau &\equiv \mathbb{E}[Y_{i1}(1) - Y_{i1}(0)|D_{it} = 1] \\ &\equiv \mathbb{E}[Y_{i1}(1) - Y_{i1}(0)|D_i = 1] \\ &= \mathbb{E}[Y_{i1}(1)|D_i = 1] - \mathbb{E}[Y_{i1}(0)|D_i = 1]\end{aligned}$$

- We can not identify τ from the CEF matrix, because we don't know the counterfactual potential outcomes.
- But we can identify τ by assuming the **parallel trends'** assumption:

$$\mathbb{E}[Y_{i1}(0) - Y_{i1}(0)|D_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i1}(0)|D_i = 0]$$

Modeling counterfactual

Additional concepts

$$\begin{aligned}\mathbb{E}[Y_{i0}(0)|D_i = 1] &= \mathbb{E}[Y_{i1}(0)|D_i = 1] \\ &\quad + \mathbb{E}[Y_{i1}(0) - Y_{i0}(0)|D_i = 0] \\ &= \mathbb{E}[Y_{i0}|D_i = 1] + \mathbb{E}[Y_{i1}(0) - Y_{i0}(0)|D_i = 0]\end{aligned}$$

So we can identify the τ :

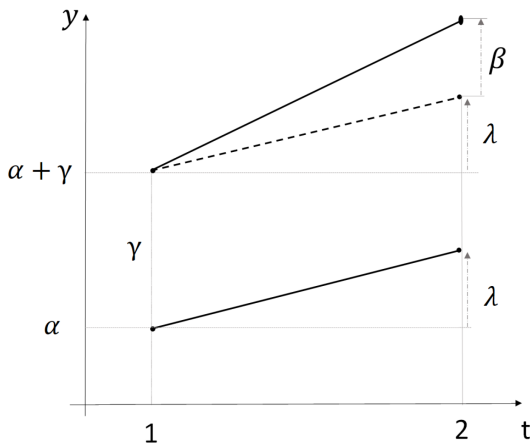
$$\begin{aligned}\tau &= \mathbb{E}[Y_{i1}(1)|D_i = 1] - \mathbb{E}[Y_{i1}(0)|D_i = 1] \\ &= \mathbb{E}[Y_{i1} - Y_{i0}|D_i = 1] - \mathbb{E}[Y_{i1} - Y_{i0}|D_i = 0]\end{aligned}$$

So we can identify the τ by comparing the difference between the treated and control group before and after the treatment.

The main assumption

The heart of the DD is the **parallel trends** assumption which is:

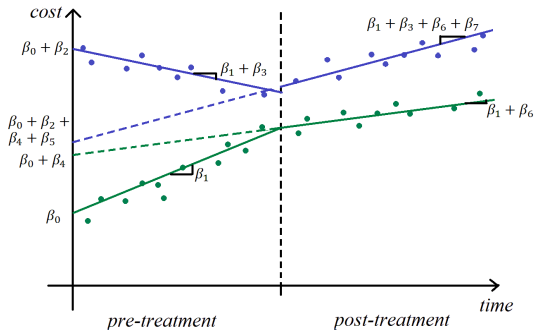
$$\mathbb{E}[y_{ist}^0|s, t] = \gamma_s + \lambda_t, \quad \mathbb{E}[y_{ist}^1|s, t] = \gamma_s + \lambda_t + \beta$$



Parallel trend assumption

- The key assumption is that the **treatment** and **control** groups would have followed the same trend in the absence of the treatment.
- This does not mean that the levels of the outcome variable are the same for the two groups.
- It is hard to test this assumption, but we can check it by plotting the outcome variable for the two groups before the treatment.
- Even if pre-trends are parallel, it is possible that other factors affect the outcome variable differently for the two groups.
- Another important assumption in DiD is exogeneity of the treatment.

Example



Additional concepts

- Dynamic Difference-in-Difference
- Triple Difference-in-Difference
- Staggered Difference-in-Difference
- Matched Difference-in-Difference
- Synthetic Control Methods

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Regression Discontinuity Design

Intuition

We learned IV, FE estimator, and Difference-in-Difference to control for unobserved heterogeneity.

The choice of the estimator depends on the research question and the data structure.

Keep in mind that all the hard works is to find a proper control group to compare the treated group with.

- Another possible way to control for unobserved heterogeneity is **Regression Discontinuity Design (RDD)**.

- ▶ RDD is a design that exploits the discontinuity in the treatment assignment
- ▶ RDD is a non-experimental design
- ▶ RDD is a quasi-experimental design

RDD provides a local randomization around the cutoff point which is the cost of this method.

Basic Setup

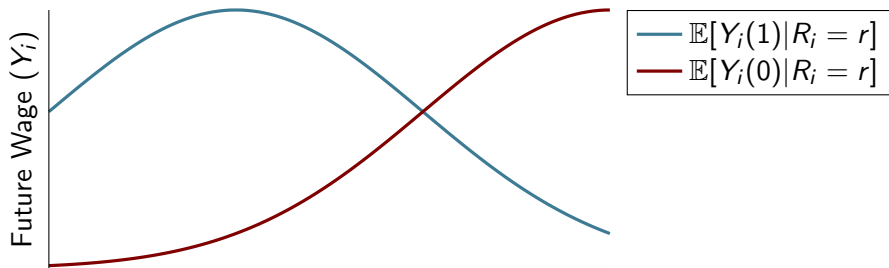
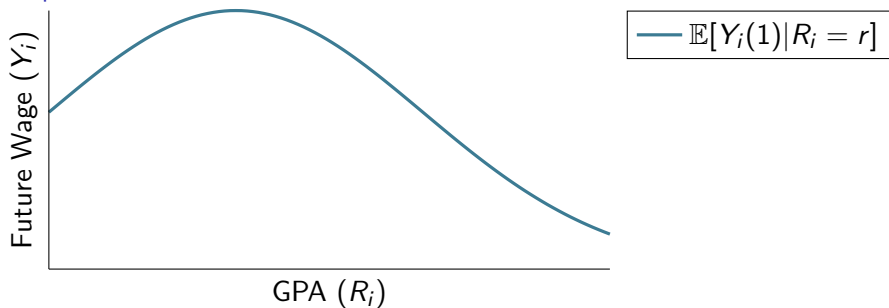
- Consider a setting where we have a treatment $D_i \in \{0, 1\}$ with POs $Y_i(0)$ and $Y_i(1)$.
- And D_i is a deterministic function of R_i (**running variable**):

$$D_i = 1[R_i > c]$$

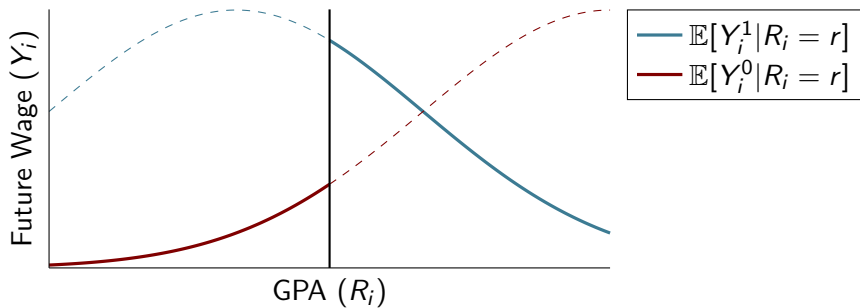
- Examples:
 - ▶ School admission test score
 - ▶ Minimum wage
 - ▶ Scholarship awarded to students above a certain GPA
- We observe $Y_i(1)$ when $R_i > c$ and $Y_i(0)$ when $R_i \leq c$.
- Basic idea:
 - ▶ Compare the outcomes of individuals just above and below the cutoff point.
 - ▶ Intuitively, treatment and control groups are similar around the cutoff point (local randomization).

Potential Outcomes

Example



Observed Outcomes



Identification

Additional concepts

- Key assumption: continuity of the potential outcomes around the cutoff point.

$$\lim_{r \rightarrow c^+} \mathbb{E}[Y_i(d)|R_i = r] = \lim_{r \rightarrow c^-} \mathbb{E}[Y_i(d)|R_i = r], \quad d \in \{0, 1\}$$

- ▶ Potential outcomes are continuous around the cutoff point.
- ▶ The population just below must not be different from the population just above the cutoff point.
- If the continuity assumption holds, we can identify the treatment effect:

$$\begin{aligned} \lim_{r \rightarrow c^+} \mathbb{E}[Y_i|R_i = r] - \lim_{r \rightarrow c^-} \mathbb{E}[Y_i|R_i = r] \\ &= \lim_{r \rightarrow c^+} \mathbb{E}[Y_i(1)|R_i = r] - \lim_{r \rightarrow c^-} \mathbb{E}[Y_i(0)|R_i = r] \\ &= \mathbb{E}[Y_i(1)|R_i = r] - \mathbb{E}[Y_i(0)|R_i = r] \\ &= \mathbb{E}[Y_i(1) - Y_i(0)|R_i = r] = \tau \end{aligned}$$

- We need to estimate the right- and left-hand side limits

$$\lim_{r \rightarrow c^+} \mathbb{E}[Y_i | R_i = r] \quad \lim_{r \rightarrow c^-} \mathbb{E}[Y_i | R_i = r]$$

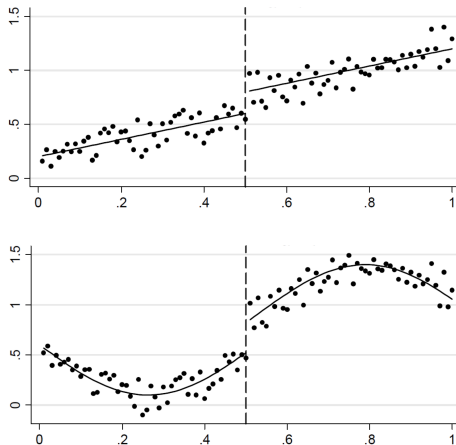
- We can estimate the limits by using the observations just above and below the cutoff point.
- We can use a linear regression model to estimate the limits:

$$Y_i = \alpha + \gamma R_i + \beta D_i + \varepsilon_i$$

- What is assumption?

RDD

Estimation



- In general, we can use nonlinear models to estimate the limits.

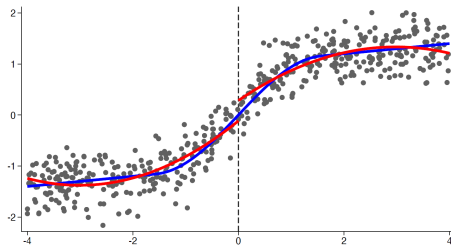
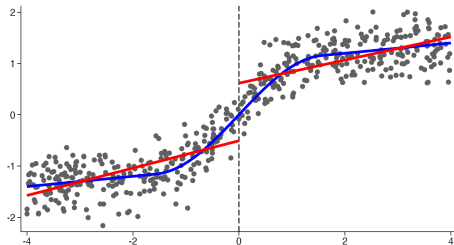
$$Y_i = f(R_i) + \beta D_i + \varepsilon_i$$

Since we extrapolate, functional form used to approximate CEF important

- Not enough to rely on OLS approximation theorems
- With insufficiently flexible specification, might mistake nonlinearity for effect
- But too flexible specification reduces precision and may be an overfitting

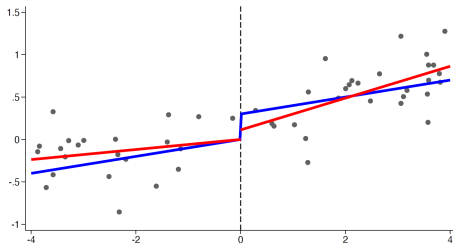
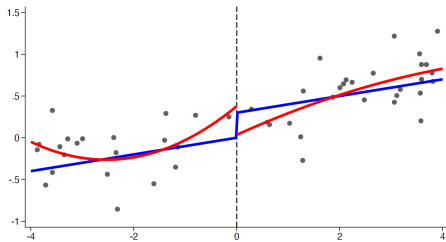
RDD

False positive



RDD

False negative



- **Global parametric:** approximate CEF with a global parametric function
 - ▶ Traditional approach
 - ▶ Can work with moderately large datasets
 - ▶ Usually uses all data, even far from cutoff
- **Local nonparametric:** approximate CEF using nonparametric techniques
 - ▶ More recent approach
 - ▶ Requires large datasets
 - ▶ Uses only small subset of data close to the cutoff

Final point

In two last methods that we cover in this lecture, visualizing the data is very important.

No one will believe you if you don't show them the data!

Assignment I

- Generate a random variable x with 1000 observations from a normal distribution with mean 0 and standard deviation 1. Also, generate a random variable $y = 2x + e$, where e is a random variable from a normal distribution with mean 0 and standard deviation 1.
- Estimate the beta coefficient of the regression of y on x .
- Now, regenerate the x with lower standard deviation and estimate the beta coefficient of the regression of y on x .
- Could you plot the average beta coefficient for different standard deviations of x ?

Assignment I

DID

- 1 This question asks you to estimate a difference in difference model. The dataset to be used is in the file "diff_in_diff_2.RData" in the folder. We have a control and treatment group observed for two time periods: one before and one after the treatment.
 - a Load the data and plot the outcome variable for the two groups before the treatment. Describe the plot.
 - b Compute the DiD estimator using the regression approach. Interpret the result.

Assignment I

Data Visualization

- ① For this question, please use the data contained in the file "industryreturns.csv". The data set contains the monthly aggregate returns (in percentage points) of 10 industries. The industries are, respectively, non-durables (1), durables (2), manufacturing (3), energy (4), hitech (5), telecoms (6), retail (7), healthcare (8), utility (9) and a category "other" for all the remaining firms (10).
 - a Convert the raw returns data in actual percentage points
 - b Examine the data. Report means by industry (e.g., 10 means). Comment on your findings.
 - c What is the volatility of the returns of the 10 industries? Comment on your findings.
 - d Standardize the returns to have mean zero and standard deviation one. Perform principal components analysis.
 - e What are the principal components' loadings?
 - f Show a scree plot of the principal components (e.g., eigenvalues on the y-axis, number of principal components on the x-axis).

Assignment II

Data Visualization

- 8 Compute the proportion of variance explained by each principal component. How many principal components do we need to explain at least 90% of the variation in the data?

Extra Assignment

Clearing the Air? The Effects of Gasoline Content Regulation on Air Quality

- ① What is the research question?
- ② What is the identification strategy? (What is the treatment and control group?)
- ③ What is your concern about the identification strategy?
- ④ What is the main result?
- ⑤ Try to replicate figure 5 and Table 2.

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Factor Models

- Factor models are a class of asset pricing models that use one or more common risk factors to explain the returns of a portfolio or a single asset.
- The idea behind the factor models is that the returns of a portfolio or a single asset are driven by the returns of the factors.
- The factors are usually portfolios of assets that are constructed based on some characteristics of the assets.
- The factor models are used to estimate the expected return of the portfolio or a single asset.

- The Capital Asset Pricing Model (CAPM) is a factor model that uses the market portfolio as the only factor.
- The CAPM is the most famous asset pricing model and it is the foundation of modern finance.
- Assumptions:
 - ① Each investor has mean-variance preferences
 - ② All investors have a one-period investment horizon and homogeneous beliefs
 - ③ All investors are price-takers
 - ④ There are no frictions such as taxes, transaction costs, etc.
 - ⑤ The risk-free asset is in zero net supply, and the risk-free rate is exogenously given

- In the CAPM, the expected return of the asset is a linear function of the expected return of the market portfolio.

$$\mathbb{E}(R_i) = R_f + \beta_i(\mathbb{E}(R_m) - R_f)$$

- Where R_f is the risk-free rate, $\mathbb{E}(R_m)$ is the expected return of the market portfolio, and β_i is the sensitivity of the asset to the market portfolio.

$$\beta_i = \frac{\sigma_{im}}{\sigma_m^2} = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

CAPM

Interpretation

- CAPM says that the expected return of the asset is the risk-free rate plus a risk premium.
- The risk premium is the product of the risk premium of the market portfolio and the sensitivity of the asset to the market portfolio.
- The sensitivity of the asset to the market portfolio is the covariance of the asset with the market portfolio divided by the variance of the market portfolio.

Multi-Factor Models

- The CAPM is a single-factor model. It was the first model to use a risk factor to explain the expected return of the assets.
- The CAPM is a good model, but it has some issues.
- The basic intuition of the CAPM still carries over to multi-factor models
- Risk premium is the sum of the products of each factor's risk premium and the sensitivity of the asset to that factor.
- The sensitivity of the asset to a factor is the covariance of the asset with that factor divided by the variance of the factor. (This holds exactly when the factors are uncorrelated; otherwise the sensitivities come from a multivariate regression.)

Multi-Factor Models

General Form

$$R_i = \alpha_i + R_f + \sum_{k=1}^K \beta_{i,k} F_k + \epsilon_i$$

- R_i is the return of the asset i
- α_i is the intercept of the regression
- R_f is the risk-free rate
- F_k is the factor k and K is the number of factors
- $\beta_{i,k}$ is the sensitivity of the asset i to the **risk-factor** k

Important Note

The factors are not firm-specific. They are common to all the firms in the market. (i.e. F_k not $F_{i,k}$)

Multi-Factor Models

- The factors are usually portfolios of assets that are constructed based on some characteristics of the assets.
- It is not necessary to use the market portfolio as one of the factors.

Examples of factors:

- Size: Small minus Big (SMB)
- Value: High minus Low (HML)
- Momentum: Winners minus Losers (WML)
- Investment: Conservative minus Aggressive (CMA)
- Profitability: Robust minus Weak (RMW)
- Term: Long-term minus short-term government bonds (TERM)
- Default: Corporate minus government bonds (DEF)
- ...

Multi-Factor Models

How to calculate the factors?

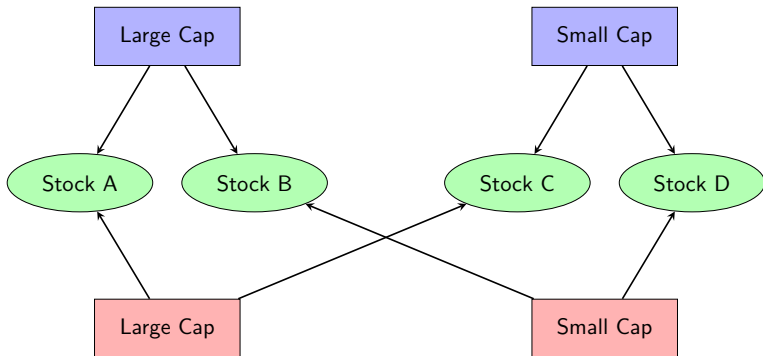
- **Sort:** Sort the assets based on the characteristics of the assets.
- **Form portfolios:** Form portfolios of assets with similar characteristics.
- **Calculate the factor:** Calculate the return of the portfolio.
- **Repeat:** Repeat the process for different characteristics.

More details at the following link: [*Construction of the Fama-French-Carhart four factors model for the Swedish Stock Market using the Finbas data*](#)

You can download the factors for the US market from [Ken French's website](#).

Factor construction

Size Example



- Blue rectangles are portfolios that are constructed based on the size of the firms at a fixed point in time.
- Red rectangles are portfolios that are constructed based on the size of the firms at a later point in time.

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Event Study

- An event study is a statistical method to assess the impact of an event on the value of a firm.
- The event is usually a corporate event such as an announcement of earnings, a merger, a new product, or a change in management. But it can be any event that is likely to have an impact on the value of the firm and is unanticipated by the market (e.g. a law change).
- Modern event studies are based on the work of Fama et al. (1969) and Ball and Brown (1968), but the idea of the event study goes back at least three decades further.
- This method is well-structured, but full of judgement calls!

Main assumption

Market is efficient and the price of the stock reflects all the available information.

Notation and definitions

- **Event:** The event should be a well-defined and well-publicized event that is likely to have an impact on the value of the firm, and the event date should be precisely defined.
 - ▶ e.g. Earnings announcement, dividend announcement, merger announcement, etc.
 - ▶ We should define the event window, which is the period of time around the event date. (e.g. 5 days before and 5 days after the event date)
 - ▶ The length of the event window will be determined by the nature of the event.

Notation and definitions

- **Return:** The return is the percentage change in the firm's stock price, including dividends, over the event window. (R_{it})
 - ▶ We can compute the return for each day in the event window or the average over the event window.
 - ▶ e.g. we might want to investigate the return for all the five days before the event up to five days after the event date.
($t = -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5$)
 - ▶ In most cases, we need to be able to separate the return of the firm from the return of the market.
 - ★ For example, if following the event date, the firm's stock price increases by 2% and the market increases by 2%, then it would be misleading to conclude that the event led to an increase in the firm's stock price.

Notation and definitions

- **Abnormal Return:** The abnormal return is the difference between the actual return and the **expected return**. (AR_{it})

$$AR_{it} = R_{it} - \mathbb{E}(R_{it})$$

- **Expected Return:** The expected return is the return that we would expect to observe **if the event had not occurred**.
 - ▶ You can see the expected return as the control group in the experiment. Our beloved counterfactual!

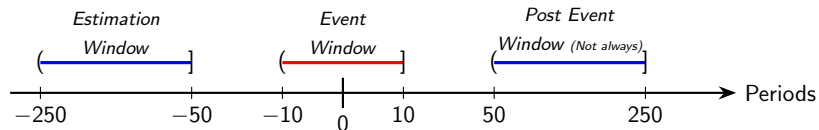
Notation and definitions

Expected Return

- There are numerous ways to estimate the expected return, but we usually use a sample of data before the event window.
- **Armitage (1995)** suggests that estimation periods can comprise anything from 100 to 300 days for daily data and 24 to 60 months for monthly data.
- Longer estimation periods are more likely to capture the true expected return, but they are also more likely to be contaminated by other events or there might be a structural change in the firm's stock price.
- There is typically a gap between the estimation window and the event window to avoid the contamination of the estimation window by the event.
- Clearly, what we would like to do is to calculate the return that would have been expected for that stock if the event had not happened at all, so that we can isolate the impact of the event from any unrelated incidents that may be occurring at the same time.

Notation and definitions

Example



Model for expected return

- **Constant Mean Return:** The constant mean return model assumes that the expected return is constant over time.
- **Market Model:** The most common model for estimating the expected return. It assumes that the return of the firm is linearly related to the return of the market. (This is a statistical model, not the CAPM.)

$$R_{it} = \alpha_i + \beta_i R_{mt} + \epsilon_{it}$$

- **Market Model with Dummy:** The market model with dummy is the same as the market model, but we add a dummy variable to capture the effect of the event.

$$R_{it} = \alpha_i + \beta_i R_{mt} + \gamma D_{it} + \epsilon_{it}$$

- **Other Models:** Use other models such as Fama-French, Carhart, etc. to estimate the expected return.
- **Control Portfolios:** Use a portfolio of firms that are similar to the firm that we are investigating to estimate the expected return.

Hypothesis Testing

- Under the null of no abnormal performance for firm i on day t during the event window, we can construct test statistics based on the standardized abnormal performance.
- The test statistic will be asymptotically normally distributed:

$$AR_{it} \sim N(0, \sigma_{AR_{it}}^2)$$

- The test statistic is:

$$\widehat{SAR}_{it} = \frac{AR_{it}}{\hat{\sigma}_{AR_{it}}} \sim N(0, 1)$$

Variance of the abnormal return

- The variance of the abnormal return is:

$$\hat{\sigma}_{u_i}^2 = \frac{1}{T-2} \sum_{t=1}^T \hat{u}_{it}^2$$

where T is the number of observations in the estimation period and $\hat{u}_{it}$ are the market-model residuals.

- An adjustment is made to the variance to reflect the errors arising from the estimation of α and β in the market model ($\hat{\sigma}_m^2$ is the variance of the market return over the estimation window):

$$\hat{\sigma}_{AR_{it}}^2 = \hat{\sigma}_{u_i}^2 \left[1 + \frac{1}{T} + \frac{(R_{mt} - \bar{R}_m)^2}{T \hat{\sigma}_m^2} \right]$$

Cumulative Abnormal Return

- It is likely that there will be quite a bit of variation of the returns across the days within the event window, with the price rising on some days and then falling on others.
- We may therefore consider computing the cumulative abnormal return over a multi-period event window by summing the abnormal returns over several periods, say from time T_1 to T_2 .

Cumulative Abnormal Return

- The cumulative abnormal return is:

$$\widehat{CAR}_i(T_1, T_2) = \sum_{t=T_1}^{T_2} \widehat{AR}_{it}$$

- The variance of the cumulative abnormal return is:

$$\hat{\sigma}_{CAR_i(T_1, T_2)}^2 = \sum_{t=T_1}^{T_2} \hat{\sigma}_{AR_{it}}^2 = (T_2 - T_1 + 1) \hat{\sigma}_{AR_{it}}^2$$

- It is common to examine
 - ▶ a pre-event window (to consider whether there is any anticipation of the event) and a post-event window
 - ▶ i.e. we sum the daily abnormal returns for a given firm i for days $t - 10$ to $t - 1$, say, and separately from $t + 1$ to $t + 10$, with the actual day of the event, t , being considered on its own.

Average Abnormal Return

So far we have covered a single firm. But what if we want to investigate the impact of the event on multiple firms?

- **Average Abnormal Return:** The average abnormal return is the average of the abnormal return of all the firms in the sample.

$$\widehat{AR}_t = \frac{1}{N} \sum_{i=1}^N \widehat{AR}_{it}, \quad \hat{\sigma}_{AR_t}^2 = \frac{1}{N^2} \sum_{i=1}^N \hat{\sigma}_{AR_{it}}^2 \Rightarrow \widehat{SAR}_t = \frac{\widehat{AR}_t}{\hat{\sigma}_{AR_t}} \sim N(0, 1)$$

- **Average Cumulative Abnormal Return:** The average cumulative abnormal return is the average of the cumulative abnormal return of all the firms in the sample.

$$\begin{aligned} \begin{cases} \widehat{CAR}(T_1, T_2) = \frac{1}{N} \sum_{i=1}^N \widehat{CAR}_i(T_1, T_2) \\ \hat{\sigma}_{CAR(T_1, T_2)}^2 = \frac{1}{N^2} \sum_{i=1}^N \hat{\sigma}_{CAR_i(T_1, T_2)}^2 \end{cases} \\ \Rightarrow \widehat{SCAR}(T_1, T_2) = \frac{\widehat{CAR}(T_1, T_2)}{\hat{\sigma}_{CAR(T_1, T_2)}} \sim N(0, 1) \end{aligned}$$

Cross-Sectional Regression

- We learned how to estimate the expected return and the abnormal return for a single firm. But what if the event has heterogeneous effects across firms?
- **Cross-Sectional Regression:** The cross-sectional regression is a regression of the abnormal return of the firm on the characteristics of the firm.

$$AR_i = \alpha + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_k X_{ik} + \epsilon_i$$

- We could use cumulative abnormal return instead of abnormal return.
- It is recommended to use heteroskedasticity-robust standard errors in the regression.

Event Study issues

Cross-Sectional Dependence

The cross-sectional regression is a common method to investigate the impact of the event on multiple firms. But there are some issues with this method.

- **Cross-Sectional Dependence:** The abnormal returns of the firms are correlated with each other. This correlation could be due to common market-wide or industry-wide shocks, and it is especially severe when the events cluster in calendar time.
 - ▶ One solution is to not use it!
 - ▶ Construct portfolios of firms having the event at the same time, and then perform the analysis on each portfolio

Small Samples and Non-Normality

- The abnormal return is asymptotically normally distributed. But in small samples, it is not normally distributed.
- I have read papers with around 200 observations that use the standard tests for hypothesis testing. But they are old!
- There are suggestions to use the bootstrap method to estimate the distribution of the test statistics or to use non-parametric tests.

Further issues

- **Variance of Returns:** The event itself raises the volatility of returns, so the variance estimated over the estimation window understates the true variance during the event window. The measured variance will be too low and the null hypothesis of no abnormal return during the event will be rejected too often.

Long Event windows

- For short periods, model discrepancies are minor. However, for longer durations, small model errors can significantly affect the calculation of abnormal returns and the event's perceived impact. Proper model design and risk allowance are crucial for long-term assessments.
- Two approaches to long event windows:
 - ① **Cumulative Abnormal Returns (CARs)**
 - ② **Buy-and-Hold Abnormal Returns (BHARs)**

$$\widehat{BHAR}_i(T_1, T_2) = \left[\prod_{t=T_1}^{T_2} (1 + R_{it}) - 1 \right] - \left[\prod_{t=T_1}^{T_2} (1 + \mathbb{E}[R_{it}]) - 1 \right]$$

- ★ When constructing BHARs the expected return is based on a non-event firm or portfolio of firms that is matched in some way to the event firm
- ★ Or you can create a portfolio of event firms and use the models that we talked about to estimate the expected return.

Event study implementation

- There are tools such as [Event Study Tools](#) for running an event study.
- But as you can see from the lecture, you as a researcher should decide about many things in the event study.
- So you should be aware of the problems and the assumptions behind the event study when you use such tools.

Assignment I

- ① Investigate the effect of Brexit on U.K. banks
 - ▶ Calculate the abnormal return for the U.K. banks after the Brexit referendum
 - ▶ Calculate the abnormal return for the U.K. banks after the first U.K. general election following the Brexit referendum
 - ▶ Calculate the abnormal return for the U.K. banks after Boris Johnson resigned as foreign secretary
 - ▶ Calculate the abnormal return for the U.K. banks after the publication of the Brexit withdrawal agreement
 - ▶ What is your conclusion?

Extra Assignment

- ① Let's calculate the abnormal return for the firm that filed a patent. We will use the WRDS to implement the event study.
 - a Download `firmEvent.csv` from Canvas. It contains the firm identifier (`permno`) and the event date (`date`).
 - b Go to [WRDS event study page](#) and answer the main questions.

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